

Portokal: Smart Workspace Console

ESP32 Based Smart Workspace Console

Student Project Presentation

PORTOKAL TEAM MEMBERS

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A Local Workspace Assistant Solves a Real Gap



THE PROBLEM

Commercial smart clocks miss local conditions

- Outdoor weather rarely reflects exact indoor workspace air quality.
- Separate devices force users to manage multiple tools.
- Not all in one.



OUR OBJECTIVES

Multi-sensor, immediate corrective feedback

- Measure local air, climate, light, motion, and proximity.
- Display readings and advice on a large touch interface.
- Add daily utility: clock, weather APIs, and audio output.

The Market Is Split Between Convenience and Sensing

CATEGORY 1

Smart Displays

Amazon Echo and Google Nest Hub offer assistants, audio, and cloud services, but do not include dedicated local air-quality hardware.

POROTAKL APPROACH

All-in-One Local Console

Combines a smart clock, localized environmental sensing, display feedback, audio capability, and external APIs.

CATEGORY 2

Air Quality Monitors

Dedicated monitors provide readings, but usually lack multimedia features, UI flexibility, AI assistant potential, and weather/time APIs.

Three-Stage Operational Workflow

STEP 1

Input Stage

- DHT11: temperature and humidity
- MQ-135: air quality changes
- LDR: ambient light
- PIR + HC-SR04: motion and distance
- Nextion touch UI, smartphone, and APIs



STEP 2

Logic

- ESP32 polls all sensors continuously.
- Threshold logic decides alerts and UI updates.



STEP 3

Output Stage

- Nextion screen renders data and menus.
- WS2812B LEDs provide visual alerts.
- Speaker plays notifications and media.
- Readings is delivered in real time.

Integrated Sensor Suite: Environment Tracking



Stable Power Distribution Protects the System



5V LOADS

Directly powered

Nextion 7-inch display, PAM8403 audio amplifier, speaker system, WS2812B LED rings, and ESP32 input power.

3.3V SENSOR RAIL

Safe sensor logic voltage

The ESP32 provides regulated 3.3V for the low-voltage modules to prevent damage and keep readings stable.

Safety step: the LM2596 was tuned with a digital multimeter to exactly 5V before connecting the screen or ESP32.

Interface and Connectivity Make Porotakl Useful Daily

TOUCHSCREEN UI

Nextion NX8048T070

Large 7.0-inch display offloads graphic rendering and communicates with the ESP32 through UART.

ONLINE DATA

Wi-Fi REST APIs

External time, location, and outdoor weather data are fetched through the ESP32 onboard Wi-Fi.

HAND GESTURE CONTROL

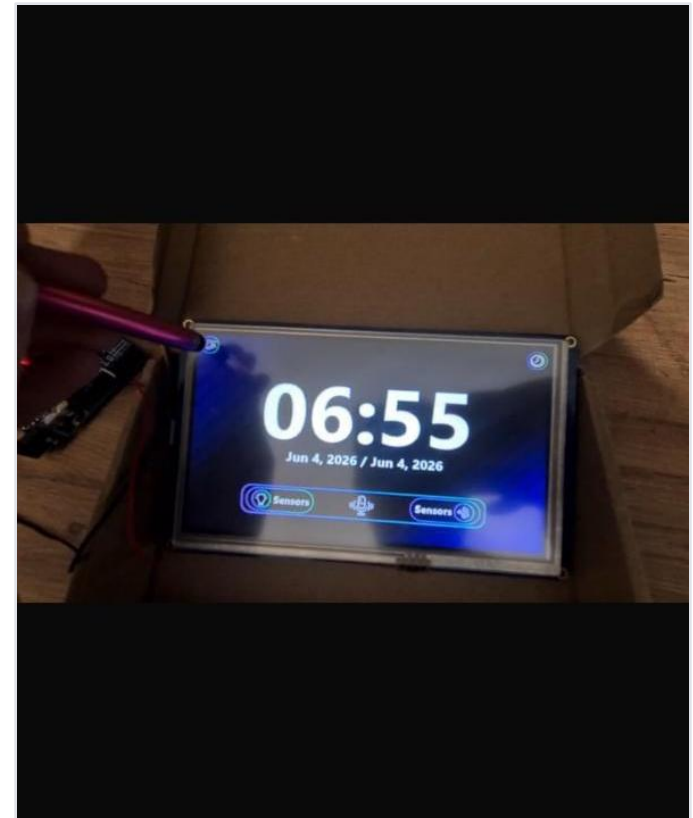
RGB LED CONTROL

Moving beyond traditional switches and smartphone apps

PHYSICAL CONTROL

Potentiometer Volume

A hardware potentiometer provides quick audio-level control without relying only on the UI.



Features

1

WATCH SYSTEM

Alarm & Time & Watch-Stop

Time is money

2

AUTO SLEEP MODE

Screen Turns off after 30s

Save Power

3

AUTO BRIGHTNESS MODE

Change Brightness of the screen according to workspace lighting

Comfortable to the eye

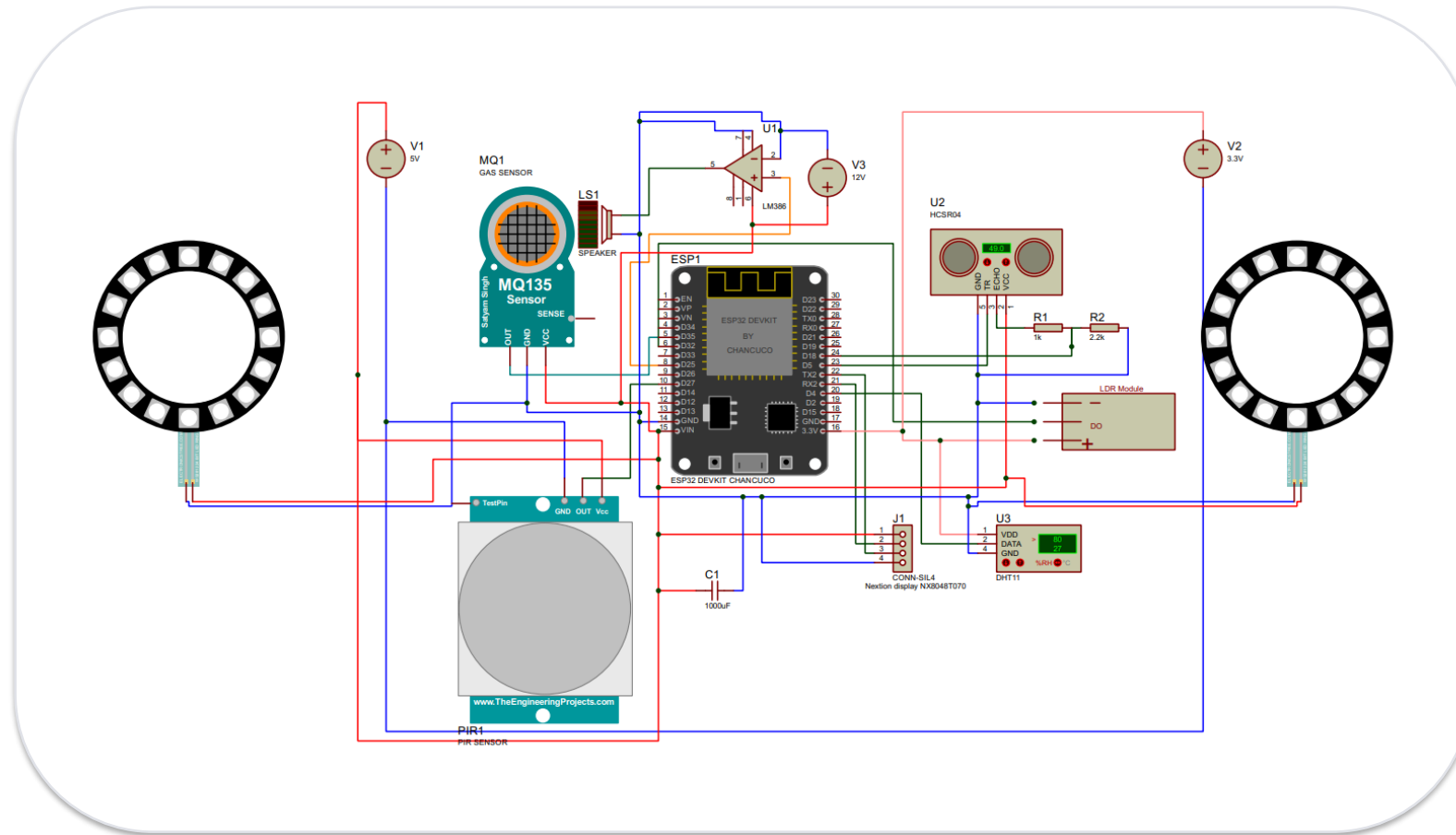
4

DUAL LANGUAGE

English & Turkish

More users = more money

Unified Electrical Architecture and Sensor Wiring



MICROCONTROLLER HUB

ESP32

Central power and data-bus integration.

SENSING

MQ, DHT, LDR, PIR

Gas, climate, ambient light, and occupancy signals.

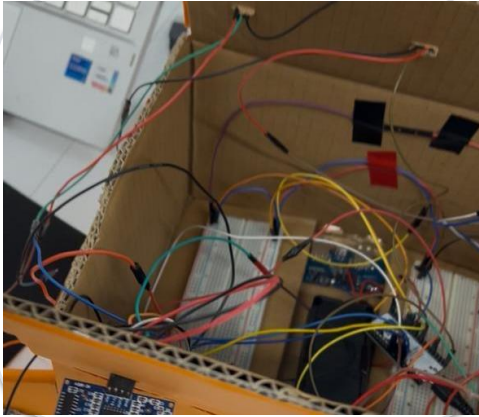
OUTPUTS

Display, LEDs, Audio

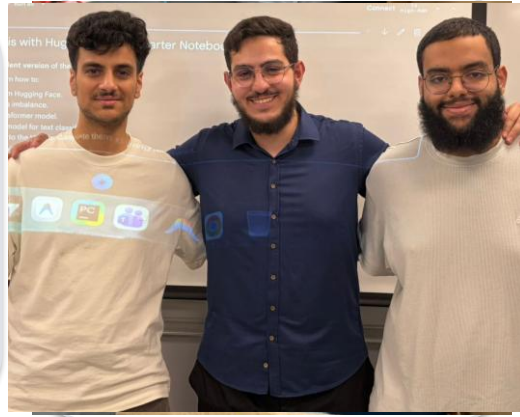
Visual and sound feedback modules for user alerts.

The Proteus circuit diagram.

Prototype Construction: From Wiring to Finished Console



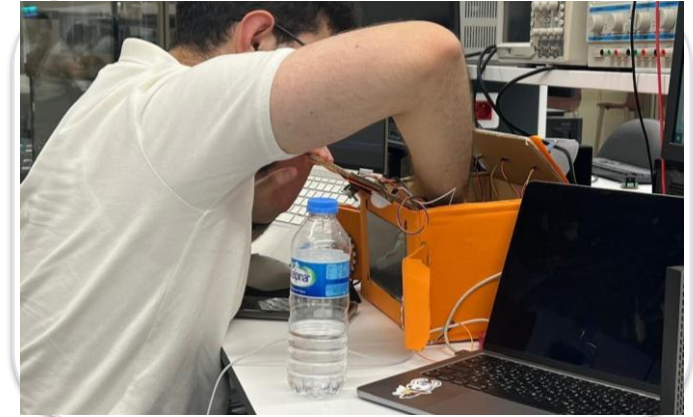
Breadboard wiring



Making brothers



Permanent solder joints



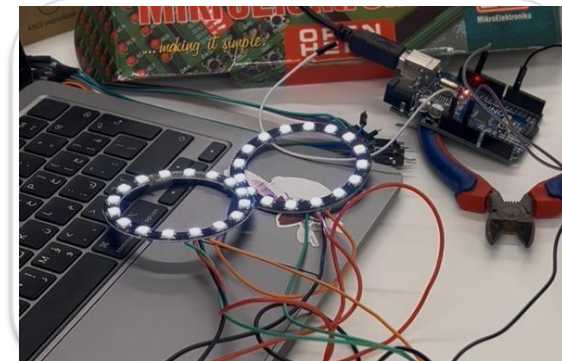
Assembly testing



Final Outer design



Circuit testing



LED/audio wiring

Work Distribution Across the Team

TEAM MEMBER 1

Emir Alammari

Documentation & Code

- Prepared report and presentation materials.
- Programmed the ESP32 logic and integration code.

TEAM MEMBER 2

Rayan Al Haiek

Power Architecture & UI

- Built power distribution and tuned the LM2596.
- Designed Nextion interface and its interaction with the other parts.
- Handled soldering and safe voltage steps.

TEAM MEMBER 3

Abdallah Moote

Build, Assembly & APIs

- Built and assembled the physical prototype.
- Wired the breadboard and connected components.
- Configured weather/time APIs and Proteus diagram.

Note: several tasks were collaborative and overlapped during construction, testing, and final integration.

Cost Proposal: Prototype Budget

TOTAL ESTIMATED COST

4,970 TL

Including screen, sensors, housing materials, and wiring.

MAIN COST DRIVER

3,500 TL

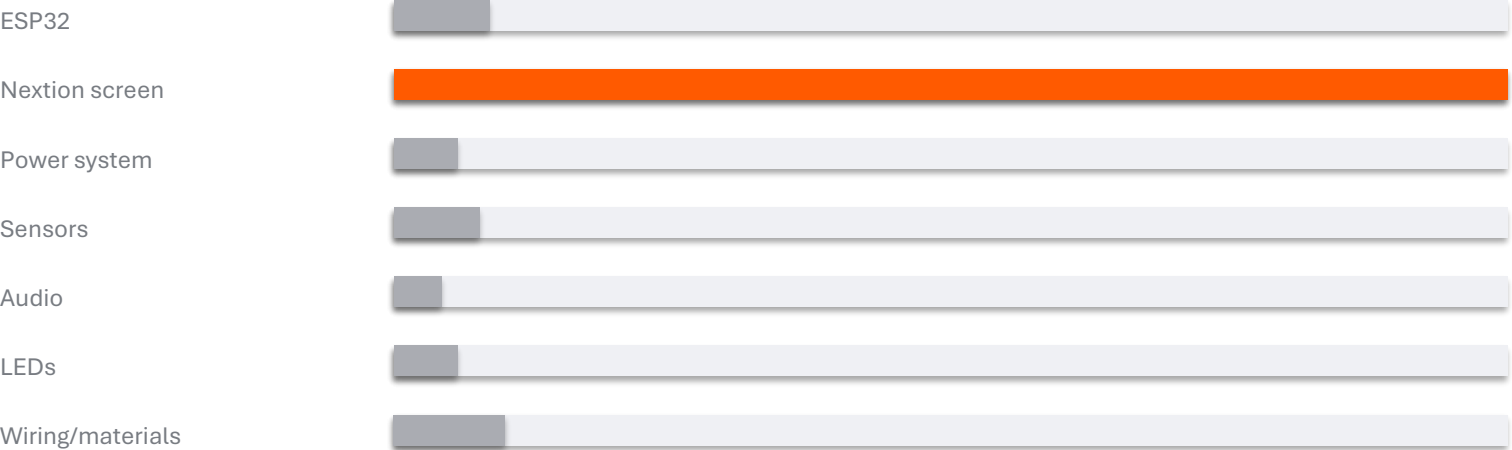
Nextion NX8048T070 7.0-inch touch display.

WITHOUT DISPLAY

1,470 TL

Estimated cost of the rest of the prototype components.

Component cost breakdown



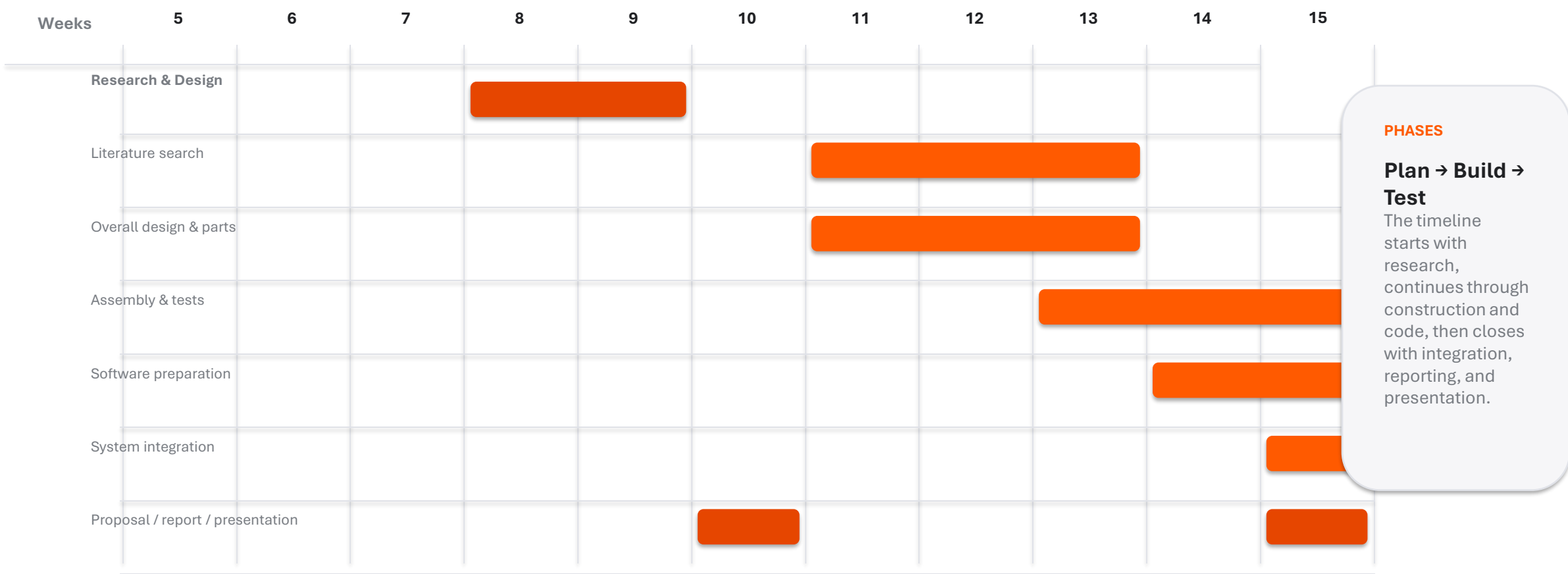
COST CONTROL

Low-cost chassis

Reused cardboard and orange craft paper reduced structural cost while matching the Porotakl visual identity.

350 TL

Project Timeline: Weeks 4–14



PHASES

Plan → Build → Test

The timeline starts with research, continues through construction and code, then closes with integration, reporting, and presentation.

Scaling Porotakl from Prototype to Product

Phase 1

Prototype Optimization

Improve sensor calibration, enclosure durability, wiring reliability, and screen interaction.

Phase 2

Cloud & Mobile Integration

Add cloud data logging and a mobile app for trend visualization and remote configuration.

Phase 3

Predictive ML Analytics

Use long-term logged data to detect patterns and generate more personalized advice.

Commercial direction: a compact desk companion that merges workspace health, productivity, visual identity, and everyday convenience.

Academic and Technical Sources

HARDWARE DATASHEETS

Core components

ESP32, Nxtion display, MQ-series gas sensors, DHT humidity/temperature sensors, LM2596 converter, and PAM8403 amplifier.

SOFTWARE RESOURCES

Development stack

Arduino IDE, DHT sensor libraries, REST API communication, and Proteus schematic design.

Thank you
<3

Hope you enjoy it as
Much as we did

Porotakl: Smart Workspace Consol
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